May 6, 2026

Matthew Kuan

Braccio Arm The Grippiest Gripper

BME203 Final Project -
Group 3b: The grippiest
Matthew Kuan, Nipon Sajib

Table of Roles

Matthew	Sajib
Programming Arduino .ino and Python .py Files	3D design including AutoCAD, Fusion360, Meshmixer or CAD
Microcontroller programming, sensor, electronics and circuit design	Verifying programmed files work well and recording demonstration
Presentation Slides 1, 2, 3, 4, 6, 7, 8, 9, 10	Presentation Slides 2, 3, 4, 5, 8, 10

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Motivation of the Design

01

Surgical Applications

- Design must support surgical applications, including gripping and manipulating tools such as scissors.
- Grippers must be precise and reliable, fitting securely onto the Braccio arm for controlled motion.
- Demonstration with a cylindrical water bottle models the ability to handle similar surgical items (e.g., alcohol bottles, tubing, vials).

02

Precise Microcontroller

- Microcontroller enables precise positioning, allowing the arm to accurately locate incision sites.
- Integrated sensors support controlled motion, ensuring stable and repeatable tool placement.
- Test-running code validates accuracy, helping refine movement paths before surgical-tool handling.

03

Sterile and Versatile

- Capable of securely grasping surgical tools and delivering them to the surgeon with controlled precision.
- Designed for versatility, allowing the grippers to handle instruments of varying shapes and sizes.
- All components are easily sterilizable, supporting safe use in surgical or clinical environments.

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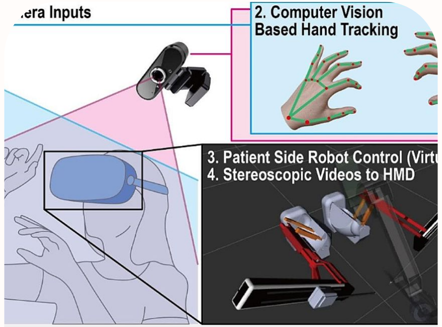
Existing Solutions and Designs



Vision-based control using MediaPipe

Systems that use MediaPipe and a standard webcam to control robotic arms [1]

MediaPipe with a Shadow Dexterous Hand, finding that the system achieves an end-to-end delay of one second, ensuring real-time usability within certain constraints [2].

Depth sensor and Leap Motion approaches

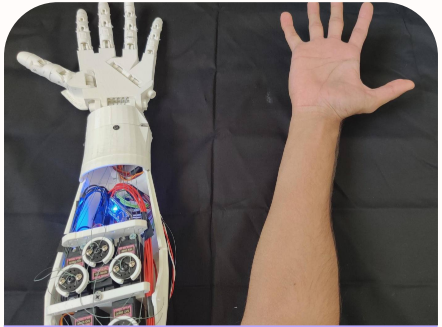
This system uses infrared sensors to track bare-hand movements in 3D space.

It mirrors these motions in real time to control the arm, enabling sterile, glove-free tool handling without complex software.

Data glove with physical sensors

Some systems place bending sensors directly on a glove worn by the operator, measuring finger curl mechanically

Optical motion capture system (OptiTrack) and a CNN+BiLSTM neural network, achieving precision in replicating hand movements [5].



EMG (muscle signal) based control

This system uses surface electrodes to translate forearm muscle contractions into robotic commands. By reading EMG signals, it detects distinct hand movements to control the arm wirelessly. While this captures intent at the muscular level, signal reliability can vary based on sweat or electrode placement.

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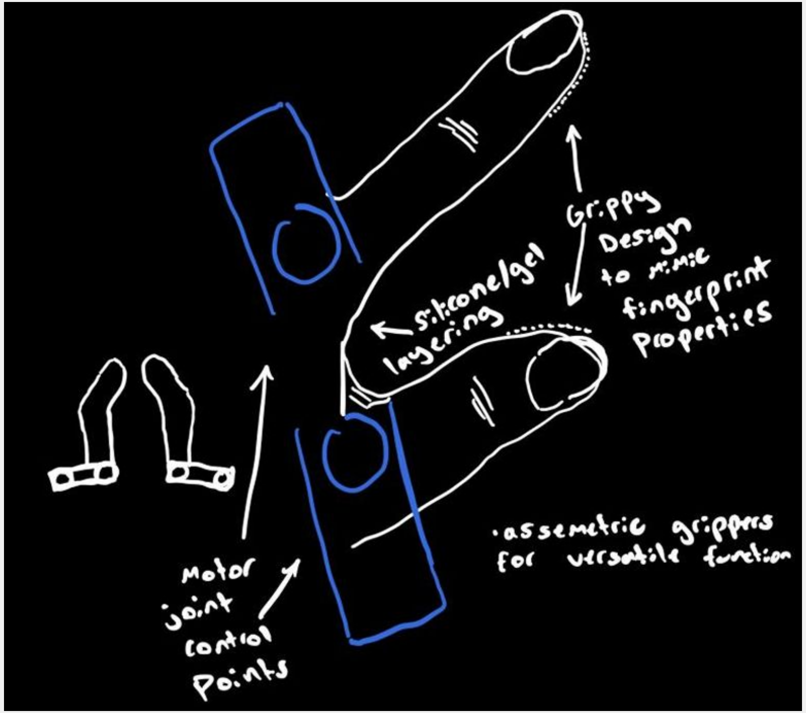
How Our Solution is Different/Improved

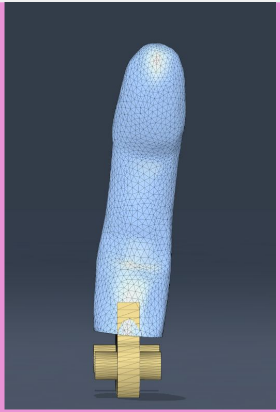


- Many competitors intend to fully cut out surgeons from surgical procedures but ours intends to be a natural extension of their intent
- Fully automated surgeries are currently suboptimal as it lacks human judgement and care [8]
- Does not require pre-programmed / verbal commands
- The use of hand like grippers mimic human hand-off dynamics, ensuring a delicate and secure grip on specialized instruments
- Needs less people in the operating room meaning:
  - More sterile environment
  - Reduced possibility for human error
  - Reduced operating room prep time
  - Reduced costs in healthcare by at least 10%

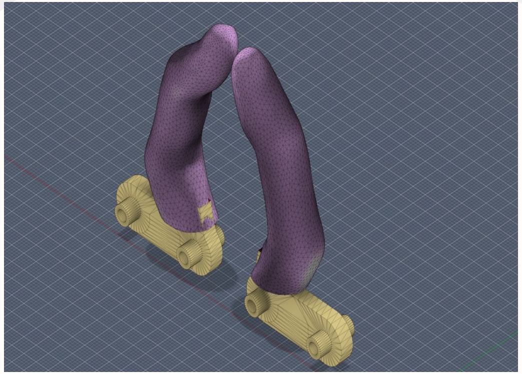
Nipon Sajib

3D Models + Intermediate Steps

- provided gripper models were used as a base we then added fingers from a borrowed stl file on thingiverse
- I Utilized plane cuts to separate the pieces and align them



- I placed the pieces on top of one another and “repaired” the mesh to get rid of the holes and empty space
- I then hit combine so the pieces would fuse

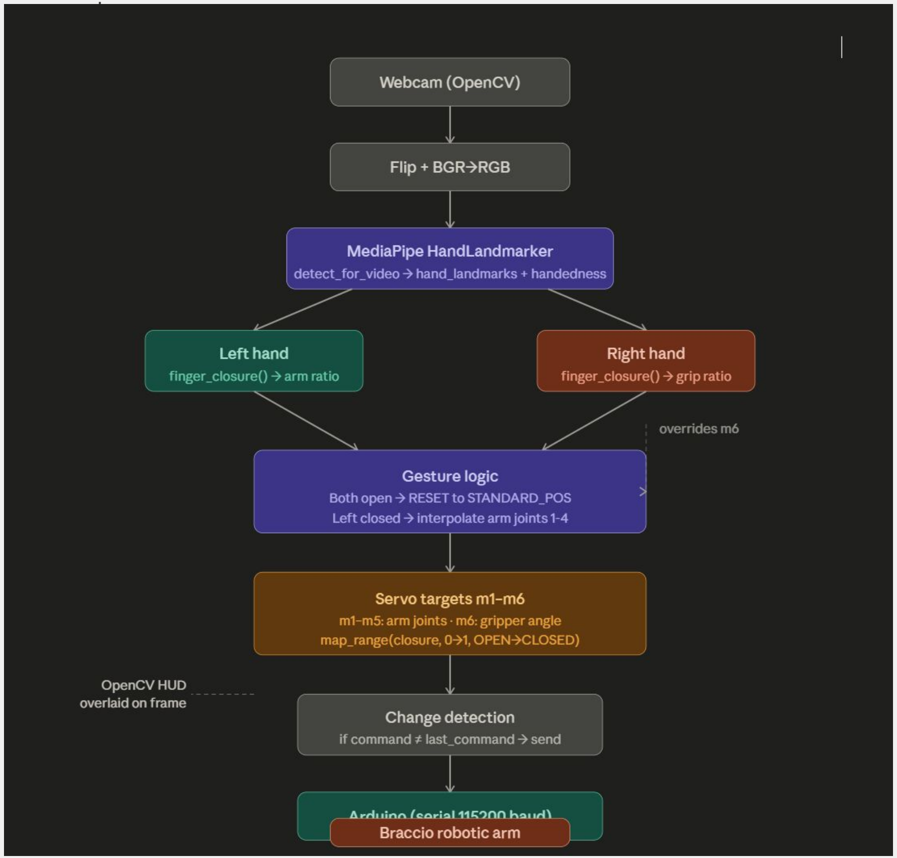


- Using the scale tool I messed around with the sizing till it was at a desired length
- I Messed with the orientations till the finger was straight and i mirrored it to further verify its alignment with the other piece

Circuits and Code

- A webcam captures a live video feed, which is mirrored so your hand movements feel natural on screen
- An AI model watches the feed and identifies where your hands are and how your fingers are positioned
- Your right hand controls the arm, closing your fist gradually angles itself to pick up an object, opening it brings it back
- Your left hand controls the gripper, squeeze to open it, close your hand to release
- If both hands are fully open at the same time, the arm resets to its default resting position
- Only changes get sent to the robot so if you hold still, no signal is transmitted and noise is reduced

Python vision control diagram



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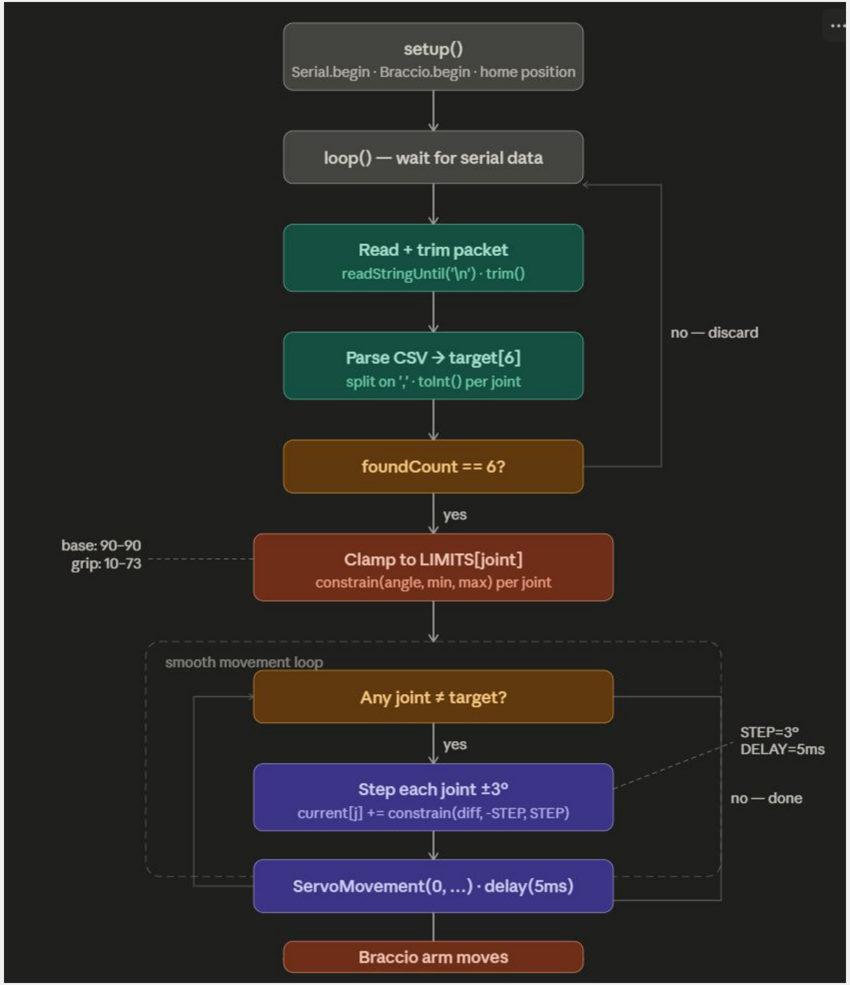
Circuits and Code

- When the Arduino powers on, it moves the arm to a known safe starting position before accepting any commands
- Then, listens over a cable connection for instructions from the laptop running the camera
- Each instruction is a simple list of six numbers: one target angle for each joint of the arm
- Before moving anything, the Arduino checks every number against a safety table and silently corrects any value that falls outside the allowed range
- Rather than snapping directly to the new position, the arm moves in tiny 3° steps with a short pause between each, making motion smooth and preventing mechanical strain
- Once all six joints reach their targets, the arm stops and waits for the next instruction

Arduino firmware diagram



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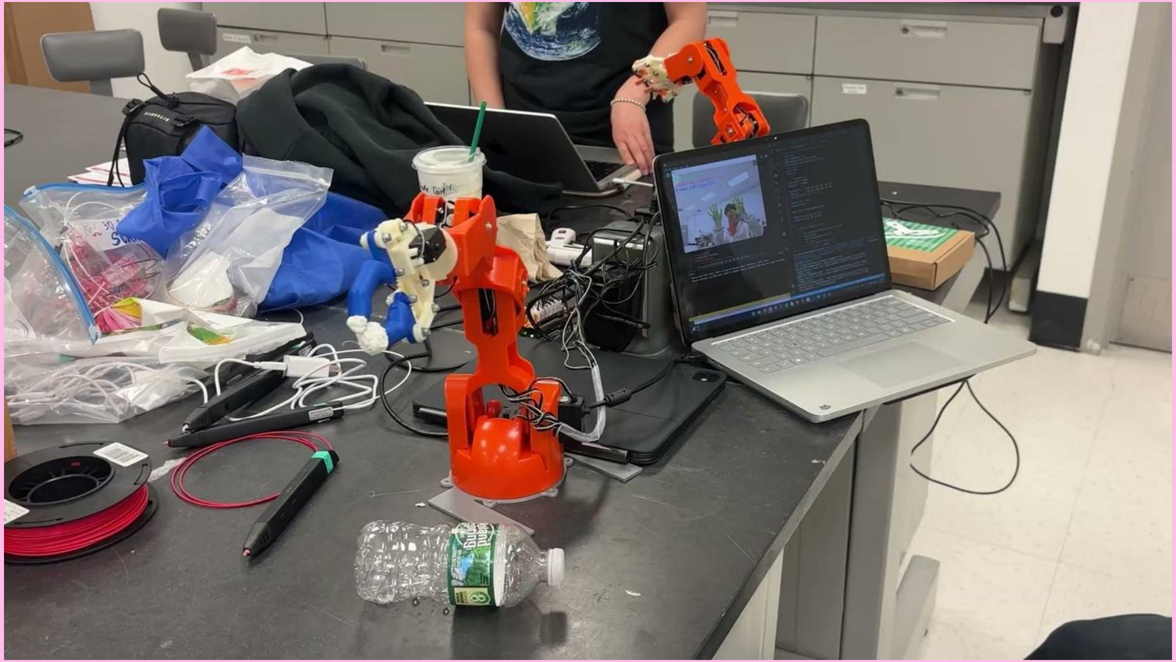

Safety Position and Demonstration

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Limitations

Finger geometry must mimic human dexterity to properly handle surgical instruments. Design accuracy is a limiting factor, requiring iterative refinement of each finger module. We will rotate design tasks and perform peer verification to minimize errors.

Challenges we faced:

- → Grippers failed to print consistently due to filament "spaghetti-ing" and print instability.
- → Initial print fractured during testing, indicating insufficient structural strength.
- → Initial print showed gripper misalignment, requiring redesign and calibration.
- → Braccio arm control required strict parameters, limiting speed and movement precision.
- → Project timeline constraints restricted the number of redesign and testing cycles.

References

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6. Hassan et al. (2018). Teleoperated Robotic Arm Movement Using EMG Signal With Wearable MYO Armband. University of Baghdad. https://arxiv.org/pdf/1810.09929
7. Vrochidou et al. (2025). Gesture-Controlled Robotic Arm for Small Assembly Lines. Machines, 13(3), 182. https://doi.org/10.3390/machines13030182
8. Wah, Jack Ng Kok. “The rise of robotics and AI-assisted surgery in modern healthcare.” Journal of robotic surgery vol. 19,1 311. 20 Jun. 2025, doi:10.1007/s11701-025-02485-0